

Gâteaux and Fréchet differentiability

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From this lecture on, we will have the opportunity to revisit with more rigorous statements some of the concepts encountered in our calculus courses.

1 Gâteaux differentiability

Definition 1.1 (Gâteaux differentiability) Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two Banach spaces over the field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let Ω be an open subset of X . The directional derivative of the function $f : \Omega \rightarrow W$ at x in the direction of $v \in V$ with $v \neq 0_V$ is

$$df(x, v) = \lim_{t \rightarrow 0} \frac{f(x + tv) - f(x)}{t} \quad \text{provided the limit exists}.$$

The notation $f_v(x)$ is also used for the directional derivative of f at x in the direction of $v \in V$. If the limit above exists for any $v \in V$ and the mapping $G_x : V \rightarrow W$ defined by

$$G_x(v) := \lim_{t \rightarrow 0} \frac{f(x + tv) - f(x)}{t} \quad (\text{GDIFF})$$

is linear and continuous, that is, $G_x \in \mathcal{L}(V, W)$, then we say that f is **Gâteaux differentiable at x** and G_x is called the **Gâteaux derivative of f at x** .

2 Fréchet differentiability

Definition 2.1 (Fréchet differentiability) Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two Banach spaces over the field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let Ω be an open subset of V . The function $f : \Omega \rightarrow W$ is Fréchet differentiable at x if there exists $F_x \in \mathcal{L}(V, W)$ (i.e., linear continuous mapping) such that

$$\lim_{h \in V, h \rightarrow 0} \frac{f(x + h) - f(x) - F_x(h)}{\|h\|_V} = 0_W. \quad (\text{FDIFF})$$

If f is Fréchet differentiable at x , then we write $Df(x) = F_x$ and call it the *Fréchet derivative of f at x* .

Remark 2.2

(i) Notice that **(FDIFF)** is equivalent to

$$\lim_{y \rightarrow x} \frac{f(y) - f(x) - F_x(y - x)}{\|y - x\|_V} = 0_W \quad \Leftrightarrow \quad \lim_{y \rightarrow x} \frac{\|f(y) - f(x) - F_x(y - x)\|_W}{\|y - x\|_V} = 0.$$

(ii) If f is Fréchet differentiable at x then the function defined by

$$\omega(y) := \begin{cases} \frac{f(y) - f(x) - Df(x)(y - x)}{\|y - x\|_V} & \text{if } y \in \Omega \setminus \{x\}, \\ 0 & \text{if } y = x. \end{cases}$$

is continuous at x with $\omega(x)$. We can then write that

$$f(y) = f(x) + Df(x)(y - x) + \|y - x\|_V \omega(y) \quad \forall y \in \Omega. \quad (*)$$

It follows from $(*)$ that

$$\underbrace{f(y) \approx f(x) + Df(x)(y - x)}_{\text{The linear approximation of } f \text{ around } x} \quad \text{for } y \text{ close enough to } x. \quad (\text{LINAPP})$$

(iii) If f is Fréchet differentiable at x then f is continuous at x . In fact, from (ii), it follows that

$$\lim_{y \rightarrow x} Df(x)(y - x) = 0 \quad \text{and} \quad \lim_{y \rightarrow x} \|y - x\|_V \omega(y) = 0.$$

Hence,

$$\lim_{y \rightarrow x} f(y) = \lim_{y \rightarrow x} [f(x) + Df(x)(y - x) + \|y - x\|_V \omega(y)] = f(x).$$

(iii) If $f : \Omega \subset V \rightarrow W$ is Fréchet differentiable at x , then f is also Gâteaux differentiable at x and $G_x = Df(x)$. In fact, for any $v \in V$, $v \neq 0$

$$\lim_{t \rightarrow 0^+} \frac{f(x + tv) - f(x)}{t} = \|v\| \lim_{t \rightarrow 0^+} \frac{f(x + tv) - f(x) - Df(x)(tv)}{\|tv\|_X} + Df(x)(v) = Df(x)(v)$$

$$\lim_{t \rightarrow 0^-} \frac{f(x + tv) - f(x)}{t} = \|v\| \lim_{t \rightarrow 0^-} \frac{f(x + tv) - f(x) - Df(x)(tv)}{-\|tv\|_X} + Df(x)(v) = Df(x)(v).$$

One could also write from $(**)$ that $f(x + tv) \approx f(x) + tDf(x)(v)$ for $|t|$ small enough and get that

$$\lim_{t \rightarrow 0} \frac{f(x + tv) - f(x)}{t} = Df(x)(v),$$

which shows that f is Gâteaux differentiable at x with $G_x = Df(x)$.

However, not every Gâteaux differentiable function is Fréchet differentiable, as shown in the example below.

Example 2.3 (Gâteaux diff. $\not\Rightarrow$ Fréchet diff.)

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) := \begin{cases} \frac{x^3 y}{x^6 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Check that the function f is Gâteaux differentiable at $(0,0)$ with $G_{(0,0)} = 0$!

The function f is not even continuous at $(0,0)$ since

$$\underbrace{\lim_{x \rightarrow 0} f(x, x^3) = \lim_{x \rightarrow 0} \frac{x^6}{2x^6} = \frac{1}{2}}_{\text{The limit of } f \text{ at } (0,0) \text{ along the curve } y = x^3} \neq 0 = f(0,0).$$

Therefore, f is not Fréchet differentiable at $(0,0)$. ■

Example 2.4 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined by

$$f(x, y) := \begin{cases} \frac{x \sin^3 y}{y^2} & \text{if } y \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Show that the function f is continuous everywhere.
- (b) Study the Gâteaux differentiability of f at the point $(1,0)$.
- (c) Study the Fréchet differentiability of f at the point $(1,0)$.

Solutions:

- (a) From the definition, it is clear that the function f is continuous at every point (x, y) with $y \neq 0$. So, it remains to study the continuity of f at every point $(a, 0)$. We want to check whether

$$\lim_{(x,y) \rightarrow (a,0)} f(x, y) = f(a, 0) = 0.$$

Using $\lim_{y \rightarrow 0} \frac{\sin y}{y} = 1$, we find that

$$\lim_{(x,y) \rightarrow (a,0)} f(x, y) = \lim_{(x,y) \rightarrow (a,0)} x \sin y \frac{\sin^2 y}{y^2} = \lim_{x \rightarrow a} x \times \lim_{y \rightarrow 0} \sin y \times \lim_{y \rightarrow 0} \frac{\sin^2 y}{y^2} = 0 = f(a, 0).$$

Hence, f is a continuous function.

- (b) Let $\vec{v} = (v_1, v_2) \in \mathbb{R}^2$ with $\vec{v} \neq (0,0)$. We first want to check whether the limit $\lim_{t \rightarrow 0} \frac{f((1,0) + t(v_1, v_2)) - f(1,0)}{t}$ exists and find it (if it does).

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{f((1,0) + t(v_1, v_2)) - f(1,0)}{t} &= \lim_{t \rightarrow 0} \frac{f((1 + tv_1, tv_2)) - f(1,0)}{t} \\ &= \begin{cases} \lim_{t \rightarrow 0} (1 + tv_1) \frac{\sin^3(tv_2)}{t^3 v_2^2} & \text{if } v_2 \neq 0, \\ 0 & \text{if } v_2 = 0 \end{cases} \\ &= \begin{cases} \lim_{t \rightarrow 0} (1 + tv_1) v_2 \left(\frac{\sin(tv_2)}{tv_2} \right)^3 & \text{if } v_2 \neq 0, \\ 0 & \text{if } v_2 = 0 \end{cases} \\ &= \begin{cases} v_2 & \text{if } v_2 \neq 0, \\ 0 & \text{if } v_2 = 0 \end{cases} \\ &= v_2 \end{aligned}$$

Hence, the function f has directional derivative $f_{\vec{v}}(1,0)$ at the point $(1,0)$ along any direction $\vec{v} = (v_1, v_2) \neq (0,0)$ with

$$f_{\vec{v}}(1,0) = v_2.$$

Clearly, the mapping $\vec{v} = (v_1, v_2) \in \mathbb{R}^2 \rightarrow f_{\vec{v}}(1,0) = v_2$ is linear and continuous. Hence, the function f is Gâteaux differentiable at the point $(1,0)$ with Gâteaux derivative

$$G_{(1,0)}(\vec{v}) := f_{\vec{v}}(1,0) = v_2 \quad \forall \vec{v} = (v_1, v_2) \in \mathbb{R}^2.$$

- (c) To check whether the function f is Fréchet differentiable at the point $(1,0)$ we need $f_x(1,0) = f_{\vec{i}}(1,0) = 0$ and $f_y(1,0) = f_{\vec{j}}(1,0) = 1$ and also we want to check whether

$$\lim_{(x,y) \rightarrow (1,0)} \frac{f(x,y) - f(1,0) - f_x(1,0)(x-1) - f_y(1,0)(y-0)}{\sqrt{(x-1)^2 + (y-0)^2}} = 0.$$

$$\begin{aligned} & \lim_{(x,y) \rightarrow (1,0)} \frac{f(x,y) - f(1,0) - f_x(1,0)(x-1) - f_y(1,0)(y-0)}{\sqrt{(x-1)^2 + (y-0)^2}} \\ &= \lim_{(x,y) \rightarrow (1,0)} \frac{x \frac{\sin^3 y}{y^2} - y}{\sqrt{(x-1)^2 + y^2}} = \lim_{(x,y) \rightarrow (1,0)} \frac{y}{\sqrt{(x-1)^2 + y^2}} \left[x \left(\frac{\sin y}{y} \right)^3 - 1 \right] \end{aligned}$$

Notice that

$$\begin{aligned} \left| \frac{y}{\sqrt{(x-1)^2 + y^2}} \left[x \left(\frac{\sin y}{y} \right)^3 - 1 \right] \right| &= \frac{|y|}{\sqrt{(x-1)^2 + y^2}} \left| x \left(\frac{\sin y}{y} \right)^3 - 1 \right| \\ &\leq \left| x \left(\frac{\sin y}{y} \right)^3 - 1 \right| \rightarrow 0 \quad \text{as } (x,y) \rightarrow (1,0). \end{aligned}$$

Hence, by the squeeze theorem, we find that

$$\lim_{(x,y) \rightarrow (1,0)} \frac{f(x,y) - f(1,0) - f_x(1,0)(x-1) - f_y(1,0)(y-0)}{\sqrt{(x-1)^2 + (y-0)^2}} = 0$$

which shows that the function f is also Fréchet differentiable at the point $(1,0)$.

Notice that (c) confirms that (b) holds. Explain why!

I still remember that many of you struggled with such questions in the second year module Advanced Calculus (2AC). I wrote the solutions with all the details in order for you to see clearly what is requested for each type of differentiability and to handle them! I suggest that you read these solutions carefully and then you redo completely all the questions.

Exercise 2.5 Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x,y) := \begin{cases} \frac{\arctan(y^2)}{\sqrt{x^2 + y^2}} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

- (a) Study the continuity of the function f .
- (b) Study the Gâteaux differentiability of the function f at the point $(0, 0)$.
- (c) Is the function f Fréchet differentiable at the point $(0, 0)$? Explain clearly your answer.

The next lecture will be on algebraic operations on Fréchet differentiable mappings such as addition, scalar multiplication and chain rule.